

Potential Energy and Conservation of Mechanical Energy

SS ILP Physics (Intermediate), Week 4 Class 1



Outline

- Conservative force and potential energy
- Gravitational potential energy – Near earth surface
- Elastic potential energy – Spring
- Conservation of mechanical energy (with conservative force)

Conservative Force

- There exist certain forces possessing a special feature that the work done by it is dependent only on initial and final positions of a motion of but not on the details of path followed.
- This type of forces (such as gravitational force, spring force) are **conservative forces**.
- A force (such as friction, air drag) that is not conservative is called a ***non-conservative force***.

R.A. Serway and J.W. Jewett, Physics for Scientists and Engineers with Modern Physics. 9E, Cengage Learning, Boston, 2014.

Conservative Force and potential energy

- We learnt that work done by conservative force $\vec{F}_c$ is path independent and depends solely on the initial and final points along the path. Thus, we can associate a potential energy only to the **conservative forces**.
- The **change in potential energy** of a body associated with a conservative force $\vec{F}_c$ is the **negative** of the work done by the conservative force in moving the body along any path connecting the initial and the final positions.

$$\Delta U = -W_c$$

- Gravitational force and the force that the spring exerts are both conservative forces – the energy “stored” as PE can be converted back to kinetic energy.

Potential Energy: Key Concepts

- Kinetic energy is associated with the motion of an object.
 - Examples: kinetic energy of linear motion, rotational kinetic energy of a rotating object (to be introduced in Term 1 – Physical World)
- **Potential energy**: Energy that is associated with the position and the configuration of a system, rather than motion.
 - Potential or possibility for work to be done
 - Examples: Gravitational potential energy, elastic potential energy of a spring, electrostatic potential energy of an electric charge (to be introduced in Term 3)

H.D. Young and R. A. Freedman, Sears and Zemansky's University Physics. 13E, Addison-Wesley, Boston, 2012.

Gravitational Potential Energy

- Gravitational potential energy is associated with a body's weight and its height above the ground.
- Work done by gravity force ($F = w = mg$) on a mass m when the mass moves vertically from y_1 to y_2 :

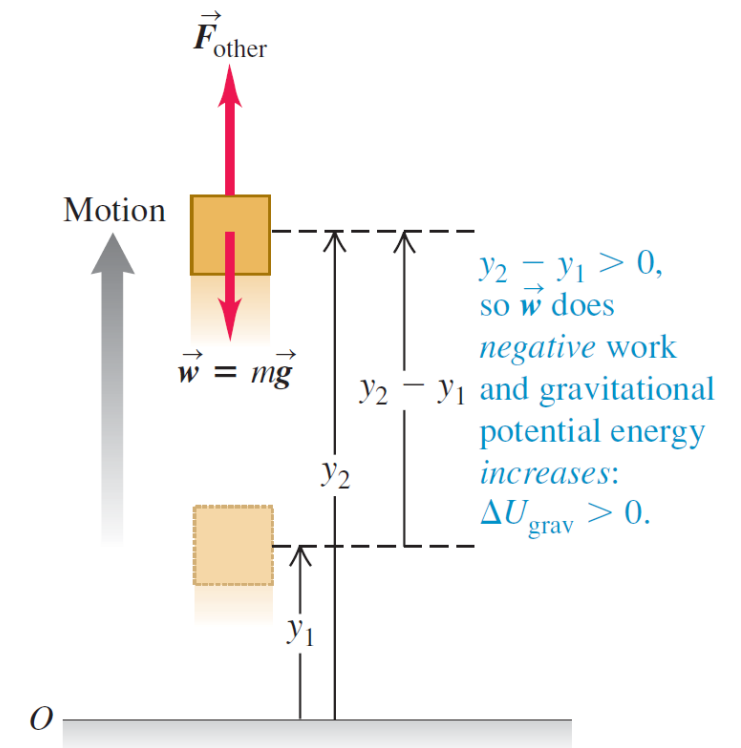
$$W_{grav} = -F\Delta r = -w\Delta y = -mg(y_2 - y_1) \\ = -(mgy_2 - mgy_1)$$

- Gravitational potential energy** is defined as

$$\Delta U_{grav} = -W_{grav}$$

$$U_{grav,2} - U_{grav,1} = mgy_2 - mgy_1$$

(b) A body moves upward



Gravitational Potential Energy

- Gravitational potential energy is associated with a body's weight and its height above the ground.
- When gravitational force does positive (negative) work, the gravitational potential energy decreases (increases).

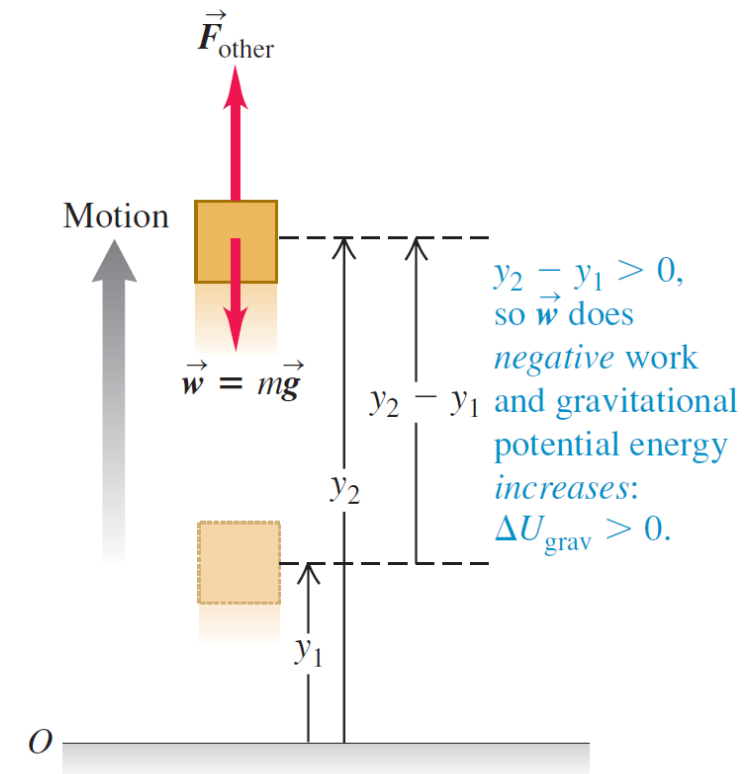
$$\Delta U_{grav} = -W_{grav}$$

$$U_{grav,2} - U_{grav,1} = mgy_2 - mgy_1$$

- Gravitational potential energy is defined as

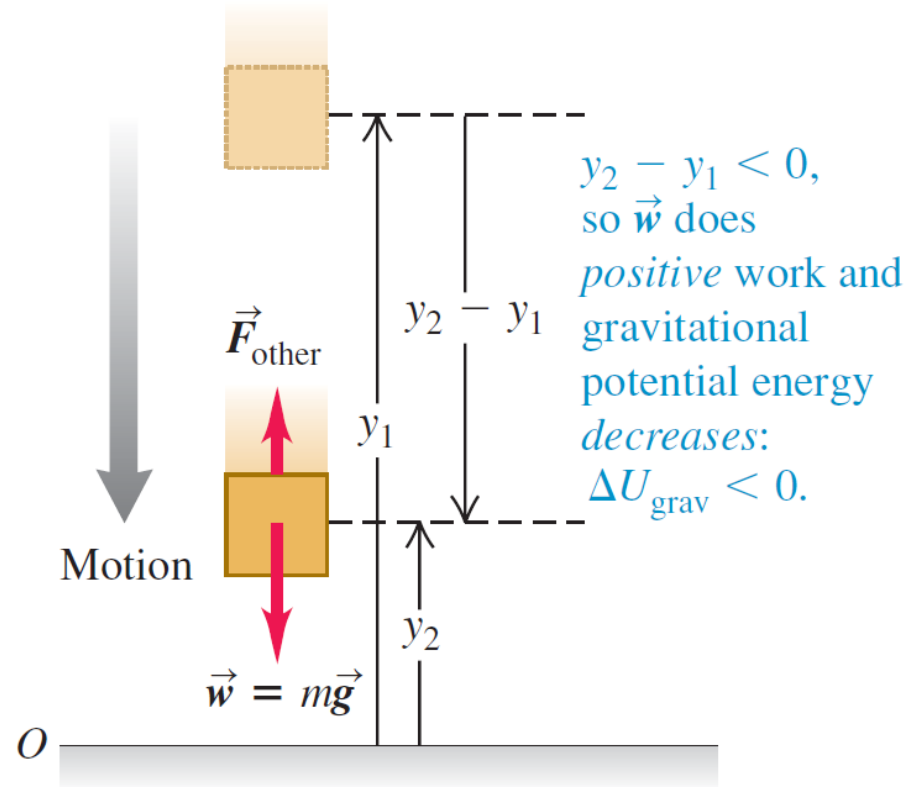
$$U_{grav} = mgy$$

(b) A body moves upward

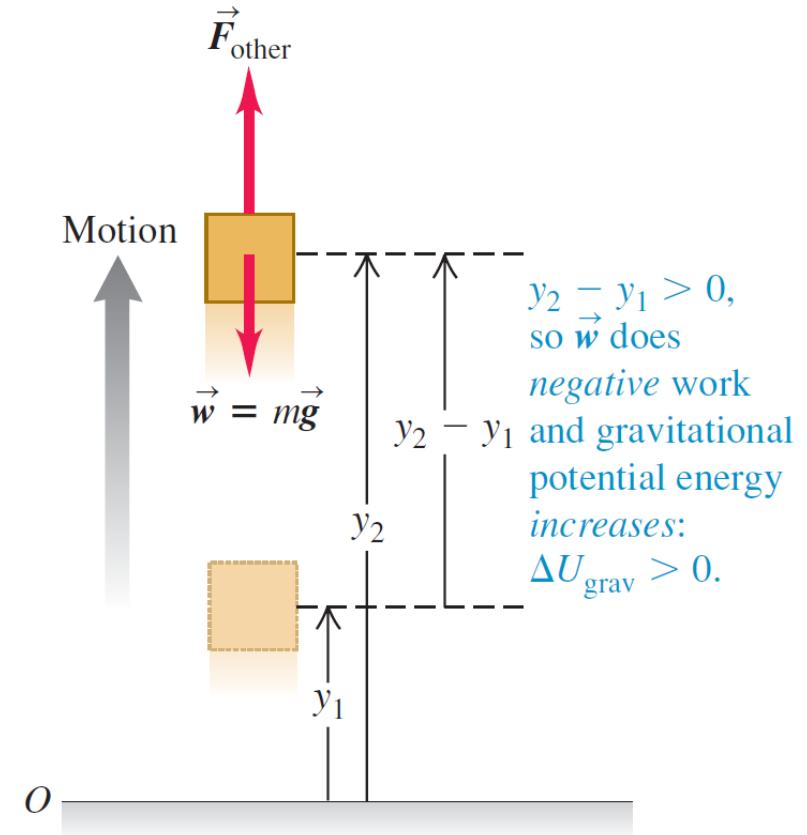


Work Done and Gravitational Potential Energy

(a) A body moves downward



(b) A body moves upward



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Concept Question

Find the gravitational potential energy of a 65-kg person on a 3.0-m-high diving board. Let $U = 0$ at the water level. Take $g = 9.81 \text{ m/s}^2$.

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Find the gravitational potential energy of a 65-kg person on a 3.0-m-high diving board. Let $U = 0$ at the water level. Take $g = 9.81 \text{ m/s}^2$.

$$U_{grav} = mgy = (65 \text{ kg})(9.81 \text{ m/s}^2)(3.0 \text{ m})$$

Elastic potential energy of a spring

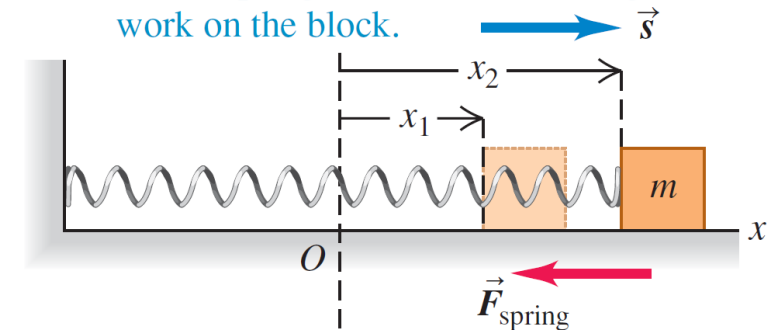
- When a deformable body, such as spring, is deformed, the work done in deforming the body is stored in the body as **elastic potential energy**.
- Consider a massless spring of spring constant k attached to a mass block m . The friction of the supporting surface is ignored.
- Work done by** the spring's **elastic force** when the mass is moved from an initial elongation x_1 to a new elongation x_2 is:

$$W_{el} = \int_{x_1}^{x_2} \vec{F}_{spring} \cdot d\vec{x} = - \int_{x_1}^{x_2} kx dx$$

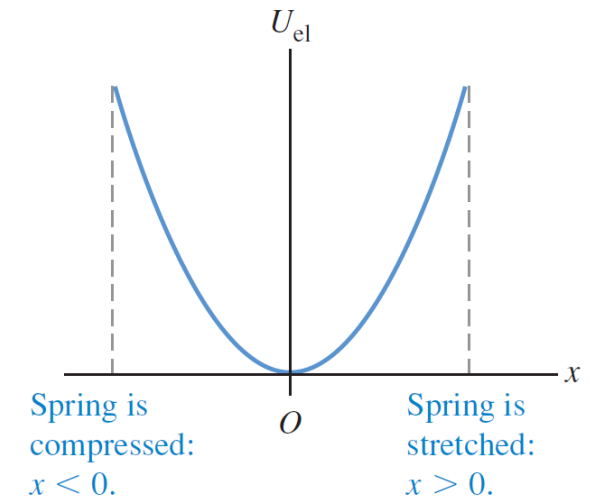
$$W_{el} = - \left[\frac{kx^2}{2} \right]_{x_1}^{x_2} = - \left(\frac{kx_2^2}{2} - \frac{kx_1^2}{2} \right) = \frac{kx_1^2}{2} - \frac{kx_2^2}{2}$$

H.D. Young and R. A. Freedman, Sears and Zemansky's University Physics. 13E, Addison-Wesley, Boston, 2012.

As the spring stretches, it does negative work on the block.



Note: A body is called **elastic** if it returns to its original size and shape after being deformed.



Elastic potential energy of a spring

- When a deformable body, such as spring, is deformed, the work done in deforming the body is stored in the body as **elastic potential energy**.
- When spring force does positive (negative) work, the elastic potential energy decreases (increases).

$$\Delta U_{el} = -W_{el}$$

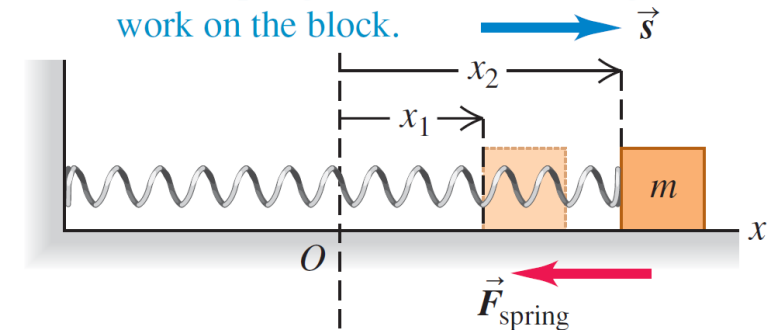
$$U_{el,2} - U_{el,1} = \frac{kx_2^2}{2} - \frac{kx_1^2}{2}$$

- Define the **elastic potential energy stored in a spring** as:

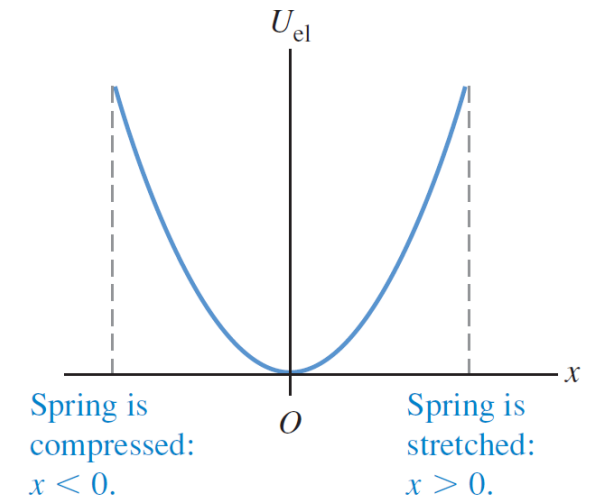
$$U_{el} = \frac{1}{2}kx^2$$

H.D. Young and R. A. Freedman, Sears and Zemansky's University Physics. 13E, Addison-Wesley, Boston, 2012.

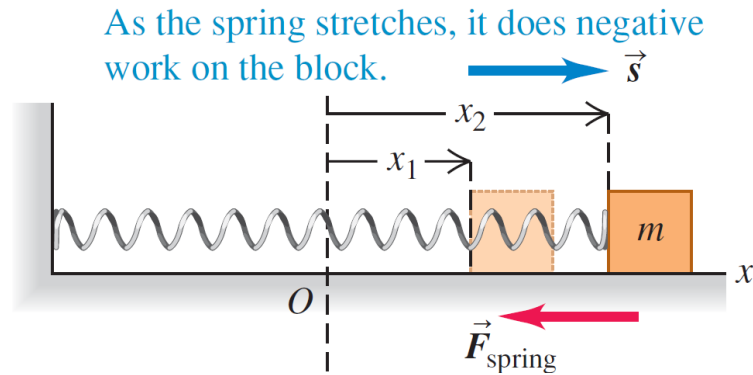
As the spring stretches, it does negative work on the block.



Note: A body is called **elastic** if it returns to its original size and shape after being deformed.

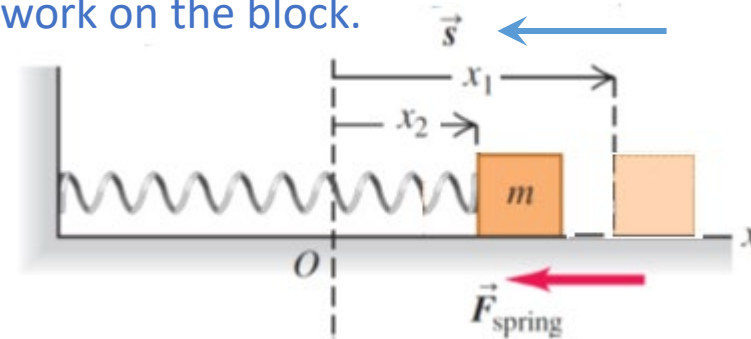


Work Done and Elastic Potential Energy



Spring force does negative work and elastic potential energy increases, $\Delta U_{el} = -W_{el}$.

As the spring compresses, it does positive work on the block.



Spring force does positive work and elastic potential energy decreases, $\Delta U_{el} = -W_{el}$.

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Concept Question

Find the potential energy of spring with constant $k = 680 \text{ N/m}$ if (a) it is stretched by 5.00 cm and (b) compressed by 5.00 cm. Let $U = 0$ at the equilibrium.

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Find the potential energy of spring with constant $k = 680 \text{ N/m}$ if (a) it is stretched by 5.00 cm and (b) compressed by 5.00 cm. Let $U = 0$ at the equilibrium.

$$(a) U_{el} = \frac{1}{2} k x^2 = \frac{1}{2} \left(680 \frac{\text{N}}{\text{m}} \right) (0.05 \text{ m})^2$$

$$(b) U_{el} = \frac{1}{2} k x^2 = \frac{1}{2} \left(680 \frac{\text{N}}{\text{m}} \right) (-0.05 \text{ m})^2$$

Conservation of Mechanical Energy

- The total mechanical energy is the summation of kinetic energy and potential energy:

$$E = K + U$$

- If the net work done is only due to conservative force(s), total mechanical energy of an isolated system of objects is constant. It means that kinetic and potential energy simply converts into one another in the system.

$$E_i = E_f \quad \longleftrightarrow \quad K_i + U_i = K_f + U_f$$

Conservation of Mechanical Energy (Gravitational forces only)

From work-energy theorem,

$$W_{total} = \Delta K$$

We obtain

$$\Delta K = W_{total} = W_{grav} = -\Delta U_{grav}$$

$$K_2 - K_1 = -(U_{grav,2} - U_{grav,1})$$

$$K_1 + U_{grav,1} = K_2 + U_{grav,2}$$

$$\frac{1}{2}mv_1^2 + mgy_1 = \frac{1}{2}mv_2^2 + mgy_2$$

Total mechanical energy
at position y_1



Total mechanical energy
at position y_2

Conservation of mechanical energy:

When only the gravity does work, the total mechanical energy of the system is conserved:

$$E = K + U_{grav} = \text{constant}$$

H.D. Young and R. A. Freedman, Sears and Zemansky's
University Physics. 13E, Addison-Wesley, Boston, 2012.

Conservation of Mechanical Energy (Spring forces only)

From work-energy theorem,

$$W_{total} = \Delta K$$

We obtain

$$\Delta K = W_{total} = W_{el} = -\Delta U_{el}$$

$$K_2 - K_1 = -(U_{el,2} - U_{el,1})$$

$$K_1 + U_{el,1} = K_2 + U_{el,2}$$

$$\frac{1}{2}mv_1^2 + \frac{1}{2}kx_1^2 = \frac{1}{2}mv_2^2 + \frac{1}{2}kx_2^2$$

Total mechanical energy
at position x_1



Total mechanical energy
at position x_2

Conservation of mechanical energy:

When only the spring force does work, the total mechanical energy of the system is conserved:

$$E = K + U_{el} = \text{constant}$$

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University Physics. 13E, Addison-Wesley, Boston, 2012.

Situations with Both Gravitational and Elastic Potential Energy

Conservation of mechanical energy
(Gravitational force)

$$E = K + U_{grav} = \text{constant}$$

Conservation of mechanical energy
(Elastic force)

$$E = K + U_{el} = \text{constant}$$

What happens when we have **both** gravitational force and elastic force?

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Conservation of Mechanical Energy: more than one type of potential energy

- Work-energy theorem still works:

$$\begin{aligned}W_{total} &= \Delta K \\W_{grav} + W_{el} &= \Delta K \\-(U_{grav,2} - U_{grav,1}) - (U_{el,2} - U_{el,1}) &= K_2 - K_1\end{aligned}$$

- Rearrange:

$$K_1 + U_{grav,1} + U_{el,1} = K_2 + U_{grav,2} + U_{el,2}$$

Conservation of Mechanical Energy

- The total mechanical energy is the summation of kinetic energy and potential energy:

$$E = K + U$$

- If the net work done is only due to conservative force(s), total mechanical energy of an isolated system of objects is constant. It means that kinetic and potential energy simply converts into one another in the system.

$$E_i = E_f \quad \longleftrightarrow \quad K_i + U_i = K_f + U_f$$

- Or, if more than one type of potential energy involved

$$K_i + \sum U_i = K_f + \sum U_f$$

Strategy for applying energy conservation

- Choose initial & final points.
- Select a zero-reference point for each form of P.E., and write the corresponding expression of P.E.
- If the net work done is only due to conservative force(s), apply

$$\boxed{E_i = E_f} \longleftrightarrow \boxed{K_i + \sum U_i = K_f + \sum U_f}$$

Example

You throw a 0.145-kg baseball straight up, giving it an initial velocity of 20.0 m/s magnitude. Find how high it goes, ignoring air resistance. Take $g = 9.8 \text{ m/s}^2$.

Example: Solution

From conservation of mechanical energy,

$$K_1 + U_1 = K_2 + U_2$$

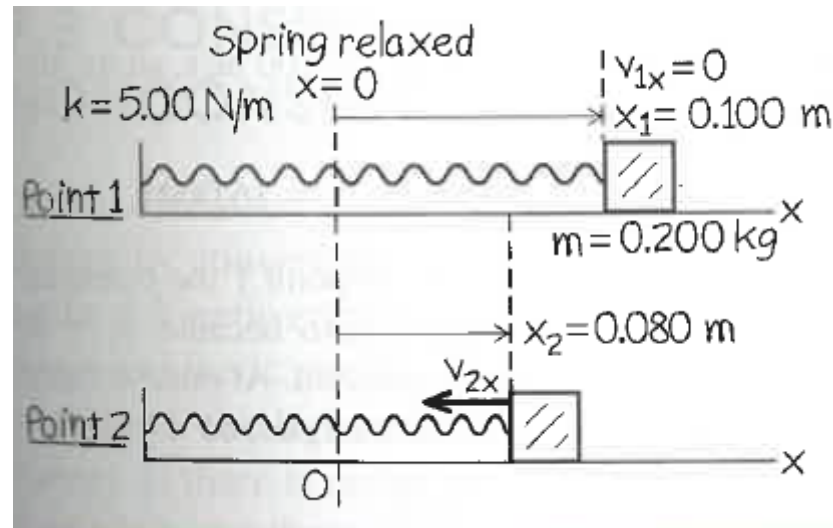
where $K_1 = \frac{1}{2}mv_0^2$ and $v_0 = 20.0 \text{ ms}^{-1}$. When the ball reaches the maximum height h , $K_2 = 0$. Hence, the conservation of mechanical energy gives

$$\frac{1}{2}mv_0^2 = mgh$$

$$h = \frac{v_0^2}{2g} = \frac{(20.0 \text{ ms}^{-1})^2}{2 \times 9.8 \text{ ms}^{-2}} = 20.4 \text{ m}$$

Case Problem 1

A glider with mass $m = 0.200$ kg sits on a frictionless horizontal air track, connected to a spring with force constant, $k = 5.00$ N/m. You pull on the glider, stretching the spring 0.100 m, and release it from rest. The glider moves back toward its equilibrium position ($x = 0$). What is its x -velocity when $x = 0.080$ m?



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Case Problem 1: Solution

The elastic potential energy the kinetic energy of the spring are

$$U_1 = \frac{1}{2} k x_1^2$$

$$K_1 = 0 \text{ (released from rest)}$$

$$U_2 = \frac{1}{2} k x_2^2$$

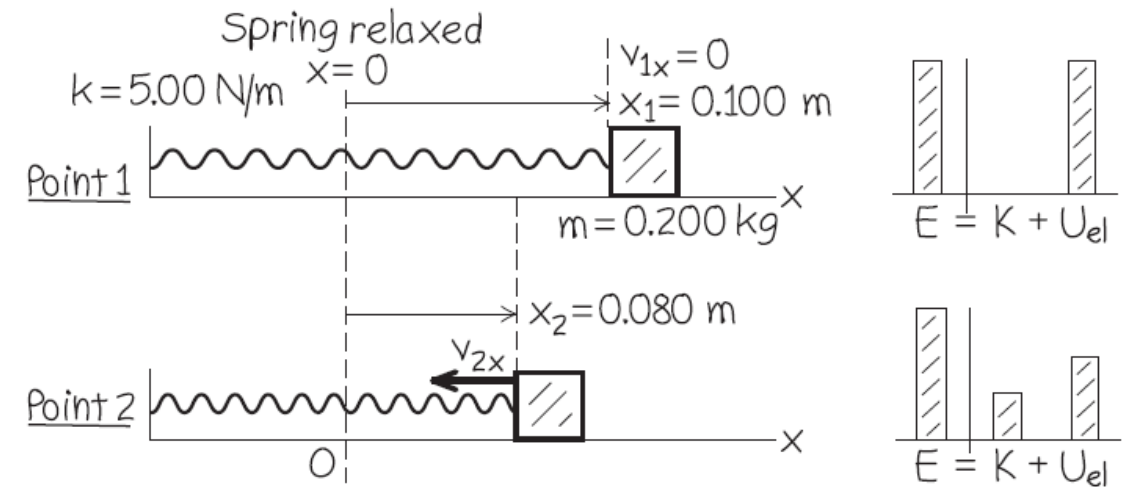
$$K_2 = \frac{1}{2} m v_2^2$$

From conservation of mechanical energy,

$$K_1 + U_1 = K_2 + U_2$$

$$\frac{1}{2} k x_1^2 = \frac{1}{2} m v_2^2 + \frac{1}{2} k x_2^2$$

$$v_2 = -\sqrt{\frac{k(x_1^2 - x_2^2)}{m}} = -0.30 \text{ m/s}$$

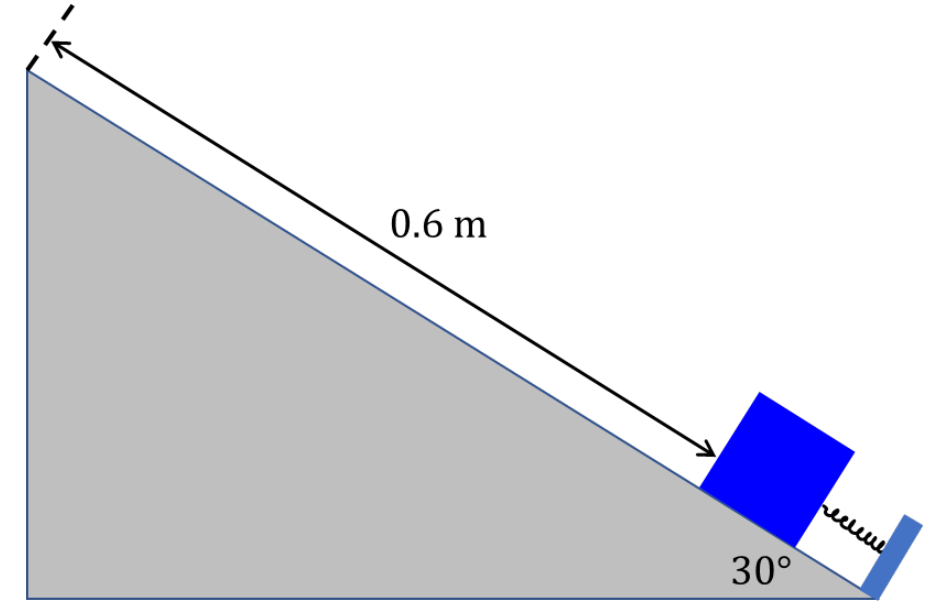


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Choose negative root because the glider is moving in negative x direction.

Case Problem 2

Consider an inclined plane with 30 degrees with the horizontal. A 0.1-kg block is held at rest and rests on a spring with spring constant 50 N/m. The spring is compressed by 0.1 m. The distance from the block to the top of the inclined plane is 0.6 m. The surface in which the block slides is frictionless. Does the block “escape” the inclined plane when it is released? Take $g = 9.8 \text{ m/s}^2$.



Case Problem 2: Solution

Set the gravitational potential energy to be zero at the initial height. The total initial energy consists only of the elastic potential energy.

$$U_{el,i} = \frac{1}{2} k(\Delta x)^2$$

Let h be the height in which the block stops. (Assume the path is sufficient.) The total final energy consists only of the gravitational potential energy.

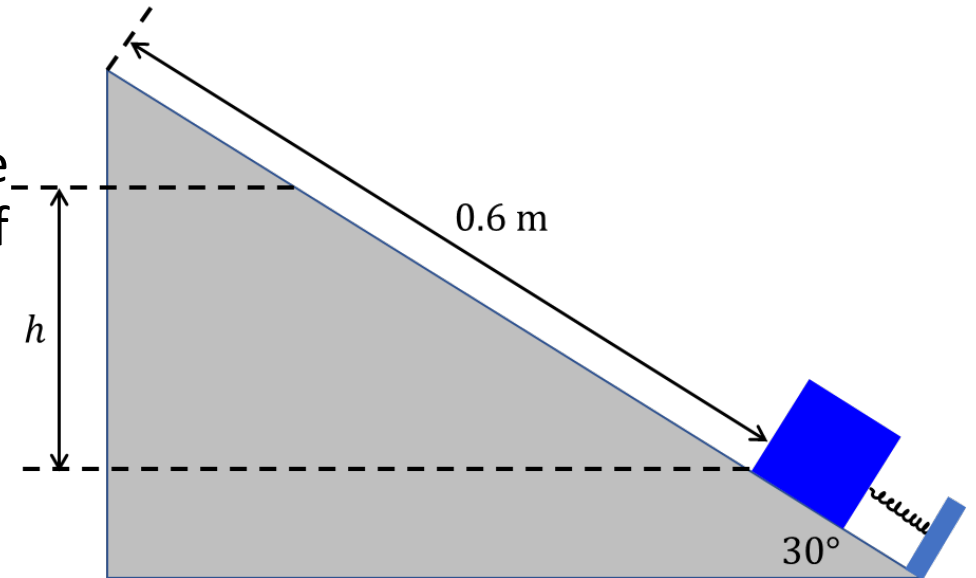
$$U_{grav,f} = mgh$$

By conservation of mechanical energy, we get

$$h = \frac{1}{2} \frac{k(\Delta x)^2}{mg}.$$

The distance moved along the inclined plane is $\frac{h}{\sin 30^\circ}$.

$$\frac{1}{2} \frac{k(\Delta x)^2}{mg \cdot \sin 30^\circ} \approx 51 \text{ cm}$$



No, there is sufficient runway for the block.

Off-class practice question(s)

Case Problem 3

Tarzan, in one tree, sights Jane in another tree. He grabs the end of a vine with length 20 m that makes an angle of 45 degrees with the vertical, steps off his tree limb, and swings down and then up to Jane's open arms. When he arrives, his vine makes an angle of 30 degrees with the vertical. Determine Tarzan's speed just before he reaches Jane. You can ignore air resistance and the mass of the vine. Take $g = 9.8 \text{ m/s}^2$.

Case Problem 3: Solution

The kinetic and gravitational potential energies of Tarzan is:

$$K_1 = \frac{1}{2}mv_1^2 = 0 \text{ (start at rest)}$$

$$U_{grav,1} = mgy_1 = mgh$$

$$K_2 = \frac{1}{2}mv_2^2$$

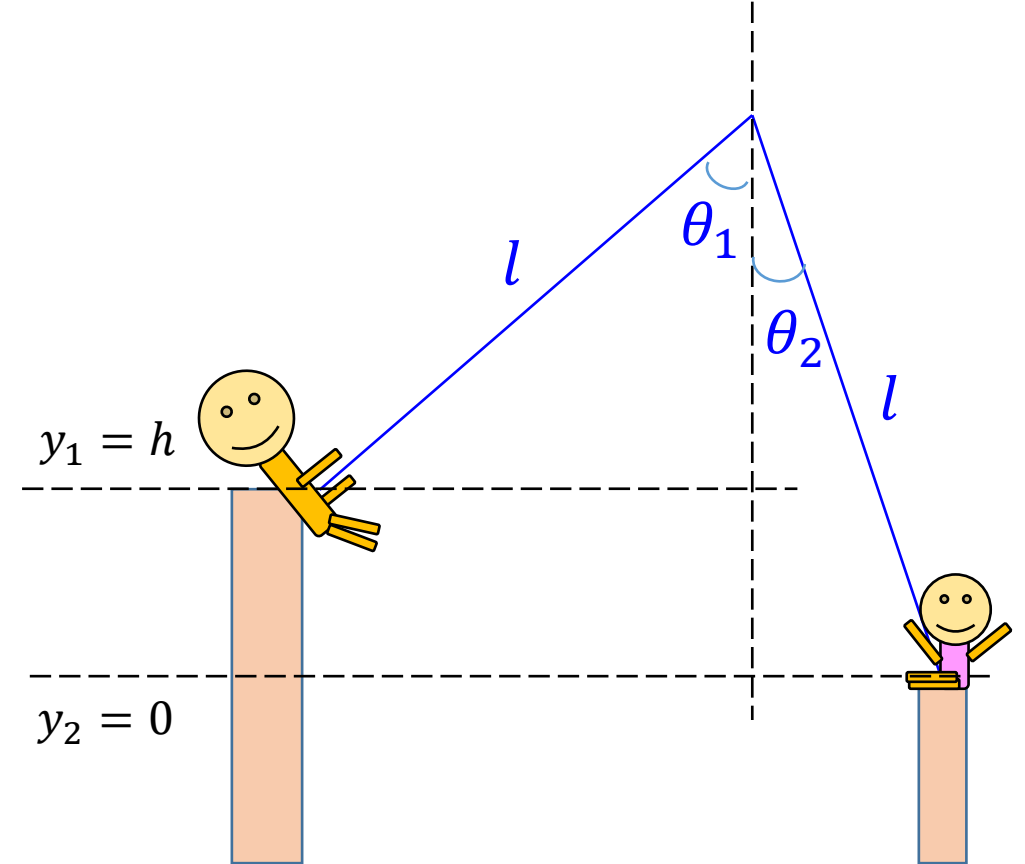
$$U_{grav,2} = mgy_2 = 0$$

By conservation of mechanical energy,

$$K_1 + U_{grav,1} = K_2 + U_{grav,2}$$

$$0 + mgh = \frac{1}{2}mv_2^2 + 0$$

$$v_2 = \sqrt{2gh}$$



From trigonometry,

$$h = l \cos \theta_2 - l \cos \theta_1$$

Hence,

$$v_2 = \sqrt{2gl(\cos \theta_2 - \cos \theta_1)} = 7.9 \text{ m/s}$$

Case Problem 4

You throw a 0.145-kg baseball straight up. Suppose your hand moves upward by 0.50 m while you are throwing the ball. The ball leaves your hand with an upward velocity of 20.0 m/s ~~(a) Find the magnitude of the force (assumed constant) that your hand exerts on the ball.~~ (b) Find the speed of the ball at a point 15.0 m above the point where it leaves your hand. Ignore air resistance. Take $g = 9.8 \text{ m/s}^2$.

Case Problem 4: Solution

(b) Set $U_{grav,i} = 0$, and $U_{grav,f} = mgh$ where $h = 15 \text{ m}$.

By conservation of mechanical energy,

$$\begin{aligned} K_i + U_i &= K_f + U_f \\ \frac{1}{2}mv_i^2 &= \frac{1}{2}mv_f^2 + mgh \end{aligned}$$

$$v_f = \sqrt{2\left(\frac{v_i^2}{2} - gh\right)} = \sqrt{2\left(\frac{20^2}{2} - 9.8 \times 15\right)} \text{ ms}^{-1} = 10.3 \text{ ms}^{-1}$$